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STC Data Analyst Project -Week 2 Report

Dataset Overview

Four datasets from Mas were provided for this week's task. The first, the **Patients** dataset, contained 999 rows (974, excluding 25 blank rows) and 26 columns (20, excluding blank columns) capturing patient demographics, geographic information, and identity data. The second, the **Procedures** dataset, was the largest of the four with 47,703 rows (47,701, excluding 2 blanks) and 11 columns (9, excluding blanks) , holding records of medical procedures performed along with associated costs and timing. The third, the **Payers** dataset, was a small reference table of 10 rows and 7 columns describing the insurance providers involved in patient billing. The fourth, the **Encounters** dataset, contained 27,893 rows (27,891,excluding blanks) and 16 columns (14, excluding blanks), documenting individual patient visits, encounter classifications, and billing details including total claim costs and payer coverage.

Together, these datasets represent a health record data for patients located in Massachusetts, spanning clinical activity from 2011 onwards.

Tool Used

The tool used for this task was Google sheets.

Dataset Errors

A thorough review of all four datasets revealed several recurring categories of data quality issues. Missing values appeared across multiple fields, including ZIP codes, reason codes, marital status, and death dates. Text fields were inconsistently formatted, with gender stored as single-letter codes ('F' and 'M') and race and ethnicity values recorded entirely in lowercase. Patient name fields in the raw data contained numeric suffixes embedded in the first last, and maiden names names, for example 'Nikita578' and 'Erdman779', making them unsuitable for display or record matching.

Date and time information across the procedures and encounters datasets was stored as combined ISO 8601 strings, which limited filtering and duration calculations. In the encounters dataset, reason codes were absent in 77% of procedure records and 70% of encounter records. The payers dataset also included a NO_INSURANCE sentinel row with several blank administrative fields including address, city, state, ZIP code, and phone number.

Data Cleaning Steps

Firstly, all datasets were explored using filter and all empty rows/columns were deleted to ensure dataset integrity. Below are the steps follow for each dataset:

Patients Dataset

Find and Replace was used to standardize **Gender** and **Marital** values from single-letter codes to full descriptive words, changing ‘F’ to ‘Female’, ‘M’ to ‘Male’, and ‘M’ to ‘Married’, ‘S’ to ‘Single’, respectively, it was also applied to the **Race** and **Ethnicity** columns to convert values from‘white’ to ‘White’ and ‘nonhispanic’ to ‘Non-Hispanic’,to improve readability and consistency. Regexplace function was used to remove symbols and numbers from the **First, Last, And Maiden** name columns, after Proper function was used to change their respective cases, and a new **Full Name** column was created by combining all name fields including the **Prefix,Suffix** columns, concatenate formula was used to achieve this. Substitute and trim were used together in a nested function to segment the addresses within the **Birth Place** column by commas in improve clarity, while,Substitute alone, was used to segment the apartment/unit numbers within the **Address** column. **Death date** nulls were retained as they correctly represent living patients, and a derived **Life Status** column was added using IF formula to simplify filtering. The 142 missing **ZIP codes** were filled with ‘NULL’, to avoid introducing false geographic information.

Procedures Dataset

The **Start** and **Stop** columns, originally stored as ISO 8601 timestamp strings, were segmented using Find and Replace and their formats changed to enable proper calculation. A **Length Of Stay** column was then calculated as the duration between both columns to facilitate time-based analysis. The 36,945 null values in **Reasoncode** and **Reasondescription** were replaced with ‘0’ and ‘Unspecified’ respectively, to maintain record completeness without fabricating clinical information. Additionally, bracketed data was extracted using ‘Split Text into columns’ and removed to standardize data for analysis.

Encounters Dataset

Proper function was used to used to change the cases of values in the **Encounterclass** column and Find and Replace to convert ‘urgentcare’ to ‘Urgent Care’. The 19,541 null values in **Reasoncode** and **Reasondescription** were replaced with ‘0’ and ‘Unspecified’. The **Start** and **Stop** datetime strings were segmented using Find and Replace into separate date and time components to support date-based filtering and duration analysis. A **Length Of Stay** column was then calculated as the duration between both columns.**Base Encounter Cost , Total Claim and Cost Payer Coverage** columns were formatted as text to ensure proper calculation during analysis.

Payers Dataset

The NO_INSURANCE row was identified as a valid sentinel entry representing uninsured patients and was retained. Its missing address, city, state, ZIP, and phone fields were documented as not applicable rather than treated as errors. Find and Replace was used to standardize **'State_Headquarted'** values from abbreviations to complete state names to ensure clarity.

Lastly, all headers were centered and the cleaned datasets were copied and pasted as values to break formula dependencies, ensure stability and prevent future errors.

Changes Made to Column Names or Structure

In the **Patients** dataset, 'Id' was renamed to 'ID', 'BIRTHDATE' to 'BIRTH DATE', 'DEATHDATE' to 'DEATH DATE', 'LON' to 'LONGITUDE', 'LAT' to 'LATITUDE', 'MARITAL' to 'MARITAL STATUS', 'ZIP' to ZIP CODE . The separate PREFIX, FIRST, LAST, SUFFIX and MAIDEN name columns were merged into a single clean 'FULL NAME' column. A new column 'LIFE STATUS' was derived from 'DEATH DATE'.

In the **Procedures** dataset, 'START' and 'STOP' became 'START TIME' and 'STOP TIME', 'PATIENT' became 'PATIENT_ID', 'ENCOUNTER' became 'ENCOUNTER ID', 'BASE_COST' became 'BASE COST', 'REASONCODE' was expanded to 'REASON CODE', and 'REASONDESCRIPTION' was expanded to 'REASON CODE DESCRIPTION'. A new column 'LENGTH OF STAY', was derived from 'START TIME'.and 'STOP TIME.

In the **Encounters** dataset, 'ENCOUNTERCLASS' was renamed to 'ENCOUNTER CLASS', 'PATIENT' became 'PATIENT_ID', 'ORGANIZATION' became 'ORGANIZATION ID', 'PAYER' became 'PAYER ID', 'START' and 'STOP' became 'START TIME' and 'STOP TIME', and was used to create the 'LENGTH OF STAY' column

Finally, in the **Payer** dataset saw 'Id' updated to 'ID', 'ZIP' to 'ZIP CODE', and 'PHONE' to 'PHONE NUMBER' and 'STATE_HEADQUARTERED' to 'STATE'.

6. Is the Dataset Now Clean and Ready for Analysis?

Yes. All four datasets are now cleaned, consistently formatted, and ready for analysis. Column names are descriptive and uniformly styled. Missing values have been handled appropriately, either retained with context, or replaced with a label where needed. Categorical text fields use consistent casing with full descriptive values. No duplicate records were found across any dataset.

Additional Notes

I assumed that any record with a missing DEATHDATE represents a currently living patient. This was necessary to keep the dataset intact for survival rate and long-term care analysis.

For the insurance analysis, I assumed that blank PAYOR fields indicate "Uninsured" patients, which is a key factor in calculating total operational cost.